



Single European Sky ATM Research Centre (Courtesy: at-one.aero)

real-time, AI-assisted flight operations management. This improvement has enhanced passenger experience while reducing costs associated with delays and inefficient routing.

Qantas provides another compelling example through its custom-built AI flight planning system known as Constellation. The system evaluates a wide range of variables, including jet stream behaviour, wind patterns, and restricted airspace activity, to generate optimised, flight-specific routes. Constellation also monitors real-time atmospheric conditions and can reroute aircraft during flight if significant changes occur, helping to reduce fuel consumption, minimise turbulence exposure, and maintain schedule reliability.

Since deploying Constellation, Qantas has achieved an annual fuel consumption reduction of approximately 2%, translating into savings of around \$92 million per year. Beyond financial gains, these efficiencies contribute to the airline's sustainability goals by lowering CO₂ emissions and improving fuel efficiency across its fleet.

In addition to cost savings, AI-driven flight operation systems play an increasingly important role in advancing sustainable aviation practices by supporting smarter, more adaptive operational decisions.

Air traffic management and control

As international air travel continues to increase, the complexity of overseeing air traffic has expanded significantly. Managing thousands of flights within congested airspace while ensuring safety, reducing delays, and maximising fuel efficiency presents a substantial challenge. Artificial intelligence is increasingly playing a vital role in this domain, providing advanced tools that support human controllers, enhance situational awareness, and automate routine decision-making.

One of the most notable developments in this area is the deployment of Remote and Virtual Towers (RVTs). These systems replace conventional airport control towers with digital platforms that utilise AI and high-definition video surveillance. Cameras installed around

airports transmit real-time visual feeds to centralised control rooms, where AI algorithms analyse the imagery to detect aircraft movements, ground vehicles, and even foreign objects or wildfires on runways. This capability allows operators to receive early alerts regarding potential safety hazards, enabling faster and more informed responses.

RVTs are particularly valuable for smaller or remote airports where maintaining a full-time physical control tower may not be financially viable. In Europe, the Single European Sky ATM Research (SESAR) initiative is at the forefront of integrating AI into air traffic management. SESAR is exploring advanced predictive systems that employ AI and machine learning to forecast traffic patterns, identify potential congestion points, and proactively resolve conflicts in flight trajectories.

Autonomous and semi-autonomous aircraft

Artificial intelligence is emerging as a key enabler in the development of autonomous and semi-autonomous aviation technologies. These advanced aircraft rely heavily on AI algorithms for critical functions such as navigation, obstacle detection, real-time decision-making, and communication with ground systems, tasks that were traditionally performed by human pilots. As the aviation industry moves towards safer, smarter, and more efficient air travel, leading aerospace organisations are investing in AI-driven



Vahana (Courtesy: aviationweek.com)